

Amit Choudhari

Systems & Microarchitectural Security Researcher

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Research Profile

Systems security researcher and Ph.D. candidate with 5+ years of security research, building on 7+ years of low-level systems experience across product engineering and applied research. Research focuses on the hardware–software boundary, combining microarchitectural reverse engineering, Linux kernel analysis, and experimental systems design to uncover side channels and build secure execution mechanisms. Work published at USENIX Security, ACM CCS, and ESORICS.

Education

Ph.D. Student in Systems Security

Dec 2022 – Present

CISPA – Helmholtz Center for Information Security

Master of Science in Cyber-Physical Systems

Sep 2020 – Oct 2022

École Polytechnique (IP Paris) · Full Scholarship

Bachelor of Engineering (B.E.) in Electronics Engineering

Aug 2009 – May 2013

Ramrao Adik Institute of Technology (Mumbai University)

Publications & Patent Application

- [1] **Amit Choudhari**, Fabian van Rissenbeck, and Christian Rossow. *Memclave: Secure In-Memory Enclave for Untrusted Hosts*. **USENIX Security 2026**. [paper]
- [2] **Amit Choudhari**, Shorya Kumar, and Christian Rossow. *NICraft: Malicious NIC Firmware-Based Cache Side-Channel Attack*. **ESORICS 2025**. [paper]
- [3] Till Schlüter, **Amit Choudhari**, Lorenz Hetterich, Leon Trampert, Hamed Nemati, Ahmad Ibrahim, Michael Schwarz, Christian Rossow, and Nils Ole Tippenhauer. *FetchBench: Systematic Identification and Characterization of Proprietary Prefetchers*. **ACM CCS 2023**. [paper]
- [4] **Amit Choudhari**, Sylvain Guilley, and Khaled Karray. *SpecDefender: Transient Execution Attack Defender Using Performance Counters*. **ASHES 2022**. [paper]
- [5] **Amit Choudhari**, Sylvain Guilley, and Khaled Karray. *CRYScanner: Finding Cryptographic Libraries Misuse*. **NICS 2021**. [paper]
- [6] **Amit Pravin Choudhari**, Basant Kumar, and Rajesh Yadav. *Enhanced User Privacy of Sensitive Data on a User Data Input Display*. **U.S. Patent Application Publication US 2021/0303737 A1, Sep 2021**. [patent]

Research Experience

CISPA – Helmholtz Center for Information Security

Dec 2022 – Present

Ph.D. Researcher – Systems & Microarchitectural Security

Advisor: Prof. Christian Rossow

- Reverse engineer undocumented CPU microarchitectural behavior and Linux kernel mechanisms using microbenchmarks and timing analysis. Current work investigates cross-core side-channel leakage and undocumented CPU prefetcher behavior.
- Co-developed *FetchBench*, a framework for systematically identifying and characterizing undocumented hardware prefetchers across 19 ARM and x86-64 CPUs. It uncovered extrapolation and replay-based prefetchers. **ACM CCS 2023**.
- Developed *NICraft*, which uses malicious NIC firmware to exploit Linux IP reassembly and mount host cache side-channel attacks. Demonstrated keystroke-activity detection at 96% precision. **ESORICS 2025**.
- Developed *Memclave*, a software-only secure execution framework for commodity UPMEM PIM using a TPM-attested hypervisor and a trusted in-PIM runtime providing control-flow integrity and cryptographic support. **USENIX Security 2026**.
- Mentored two bachelor's thesis students on systems-security research projects.

ETH Zurich

Apr 2022 – Sep 2022

Academic Guest – Systems Security Research

Supervisor: Prof. Shweta Shinde

- Developed a TEE prototype for QEMU-based RISC-V systems to protect user-space applications from higher-privileged monitor software.

Secure-IC

May 2021 – Mar 2022

Master's Research Project – Systems & Hardware Security

Supervisor: Prof. Sylvain Guilley

- Built *SpecDefender*, a hardware performance counter (HPC)-based runtime system that detects Spectre v1/v2 activity and mitigates execution at runtime. *ASHES 2022*.
- Built *CRYScanner*, a rule-based dynamic analysis tool for detecting cryptographic-library misuse in OpenSSL/SoftHSM applications. Evaluated it on public GitHub repositories and Stack Overflow examples. *NICS 2021*.

IIT Madras, Chennai, India

Aug 2013 – Mar 2015

Research Associate – Secure Embedded Systems

Supervisor: Prof. V Kamakoti

- Developed a DRDO secure-tablet platform on i.MX6 with U-Boot secure boot, TEE, and tamper-detection mechanisms.
- Ported the MAX98089 audio-codec driver and other i.MX6 peripherals. Performed board bring-up, schematic verification, and functional validation.

Engineering Experience

Qualcomm India Pvt Ltd

Mar 2017 – Nov 2020

Senior Software Engineer – TrustZone / Platform Security

- Designed and implemented ARM TrustZone and hypervisor-based Secure Camera and Secure UI components on Snapdragon platforms for payment and authentication use cases.
- Led the Alipay face-authentication project and a two-member engineering team; designed and implemented hypervisor-based isolation for camera data and control paths and frame processing in a dedicated VM.
- Secured end-to-end capture, display, and input paths so camera frames, display buffers, and touch/keypad coordinates remained accessible only to trusted components.

embedUR Systems, Chennai, India

Apr 2015 – Feb 2017

Embedded Software Engineer – Wireless Networking

- Developed embedded networking features for wireless controllers and access points on Qualcomm Atheros (QCA) platforms.
- Enhanced and debugged QCA network drivers for stability and dynamic channel and bandwidth-assignment configurations.

Technical Expertise

Programming	C, C++, Python, Bash
Systems	Linux kernel and device drivers, embedded Linux/Android, TCP/IP networking, virtualization/hypervisors, bootloaders (U-Boot), performance analysis, system debugging
Security	Microarchitectural side channels, CPU/cache/prefetcher reverse engineering, ARM TrustZone/TEEs, secure boot, TPM attestation, firmware security, tamper detection
Platforms	x86, ARM/Qualcomm Snapdragon, RISC-V, UPMEM PIM, SmartNICs, Freescale i.MX6/QorIQ
Tools / Interfaces	GDB, Git, QEMU, JTAG, hardware performance counters, I2C, SPI, UART, PCI, USB